

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Physiology DLA 20

Topic: Tubular Fluid/Plasma Concentration
Profiles

(Notes to go with online video)

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Learning Objective

- SOM.MK.II.BPM1.3.CPR.2.PHYS.0329. Schematically illustrate the TF/P profiles for Na^+ , K^+ , Cl^- , HCO_3^- , inulin, creatinine, urea, PAH, amino acids and glucose.
- SOM.MK.II.BPM1.3.CPR.2.PHYS.0380. Predict how the presence or absence of ADH affects each TF/P profile.

Video Link:

<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a848d3ab-998b-4987-a3ca-b21f00f0b6ff>

Single Nephron Features

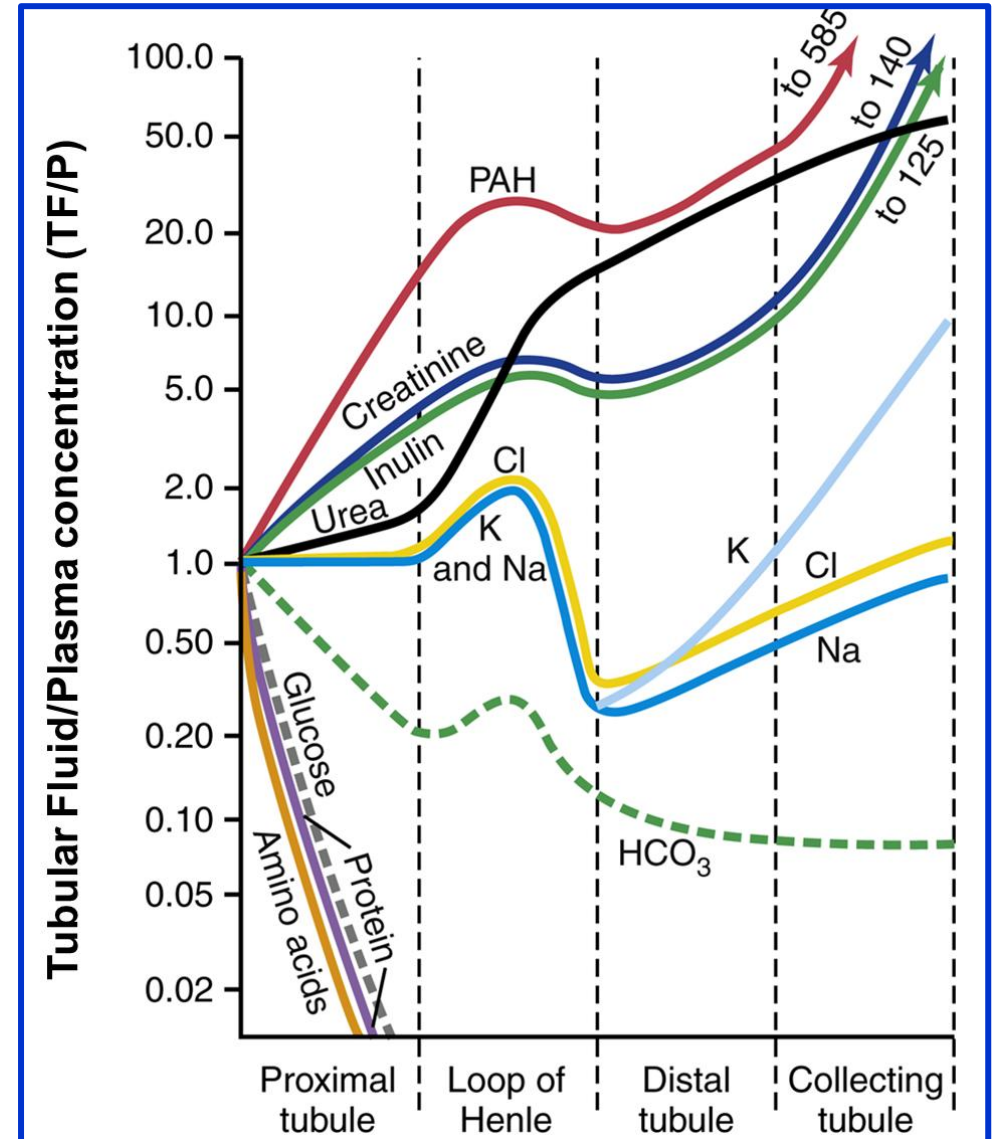
- Review tubular reabsorption of major solutes in different parts of the nephron.
- Table below shows **permeability** characteristics of sodium, chloride, water and urea in different parts of the nephron with or without ADH.

Tubular site	Active NaCl Transport	Permeability		
		H ₂ O	NaCl	Urea
Proximal tubule	++	++	+	+
Thin descending limb	0	++	+	+
Thin ascending limb	0	0	+	+
Thick ascending limb	++	0	0	0
Distal tubule	+	+ADH	0	0
Cortical collecting tubule	+	+ADH	0	0
Inner medullary collecting duct	+	+ADH	0	+ADH

ADH, antidiuretic hormone; NaCl, sodium chloride; 0, minimal level of active transport or permeability; +, moderate level of active transport or permeability; ++, high level of active transport or permeability; +ADH, permeability to water or urea is increased by ADH.

Tubular Fluid-to-Plasma [TF/P] Curves of solutes

- Glomerular ultrafiltrate concentration of all freely filtered solutes is same as plasma concentration.
- Therefore, [TF/P] ratio of all freely filtered solutes is 1.0 in the Bowman's capsule.
- TF/P ratio of all solutes begin from this point (1.0) in the figure.
- The different parts of the renal tubules are separated by the discontinuous vertical lines.



Tubular Fluid-to-Plasma concentration of a solute: [TF/P] Ratio or TF/P profile

- The [TF/P] ratio compares the concentration of a substance (solute) in tubular fluid with its concentration in plasma.
- Using micropuncture technique, the concentration of a solute can be measured at various tubular luminal sites along the nephron.
- Since plasma concentration, P in the ratio is constant, any alteration in the [TF/P] ratio or TF/P profile of a solute reflects its alteration in the tubular fluid concentration.

[TF/P] Ratio Interpretation

If [TF/P] ratio of a solute is 1.0:

The amount of solute and water reabsorption are EQUAL.

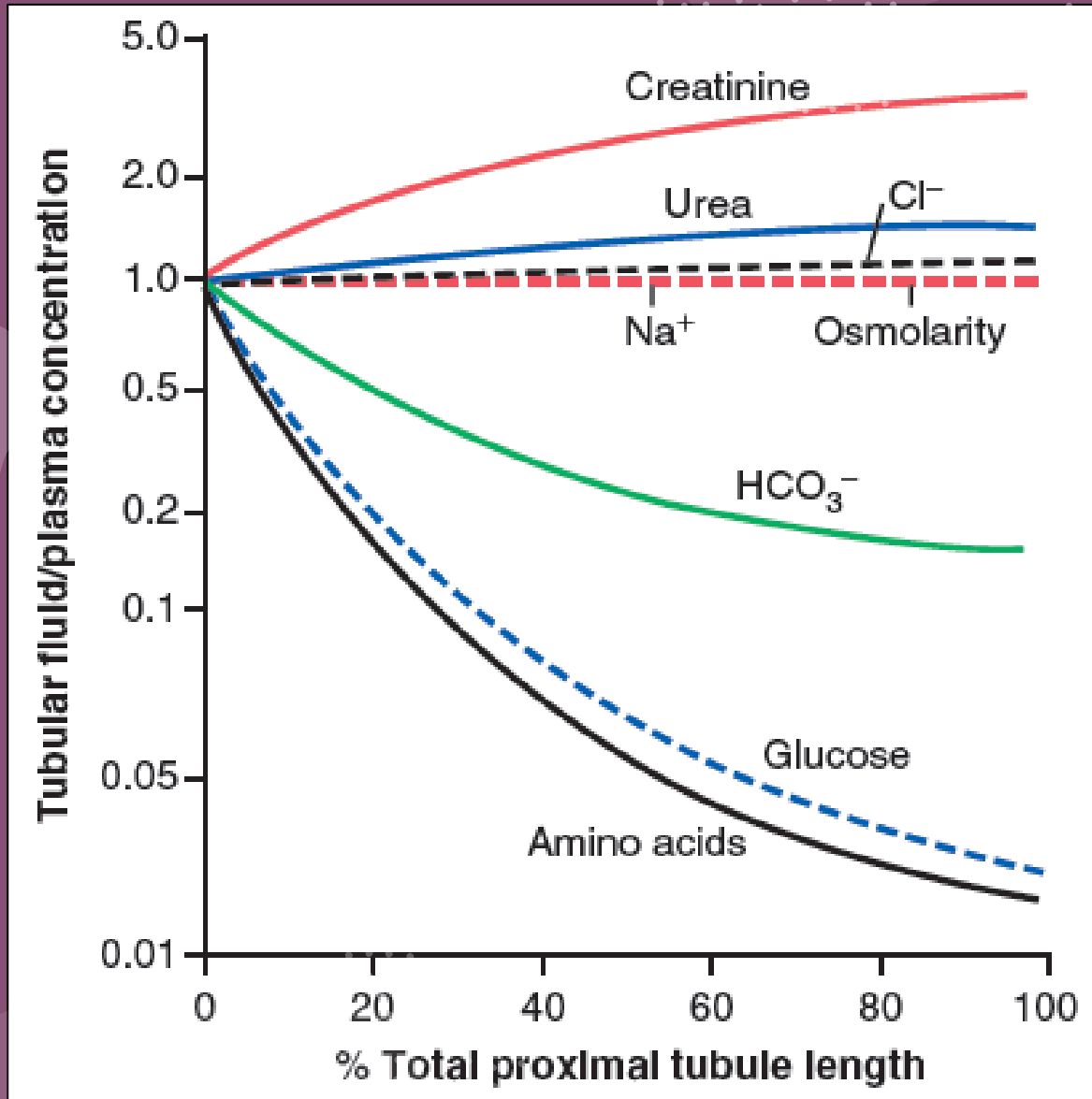
If [TF/P] ratio is less than 1.0 (<1.0):

The amount of solute reabsorbed is relatively MORE than the amount of water reabsorption.

If [TF/P] ratio is greater than 1.0 (>1.0):

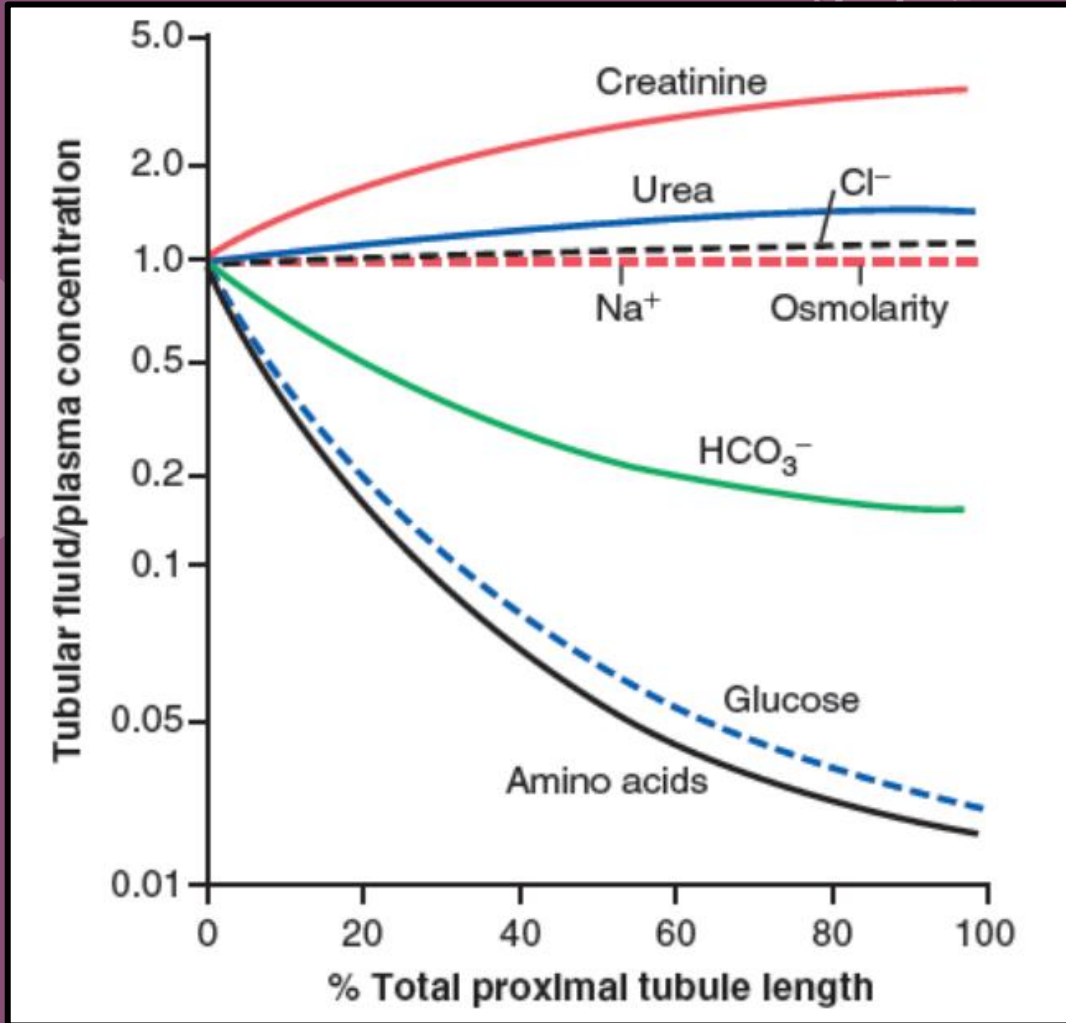
Either water reabsorption is relatively MORE than the solute reabsorption; or the solute is being secreted into the tubular lumen.

[TF/P] Curves of solutes in PCT



- [TF/P] ratio of Sodium (Na⁺) remains at 1.0 because equal amount of both sodium and water are reabsorbed in PCT.
- Glucose and Amino Acids are completely reabsorbed in the PCT, therefore their [TF/P] ratio approaches zero. (concentration of these solutes is zero in the distal end of PCT).
- Bicarbonate (HCO₃⁻) is also avidly reabsorbed, but the reabsorption is not complete in the PCT.

[TF/P] Curve of Chloride in PCT



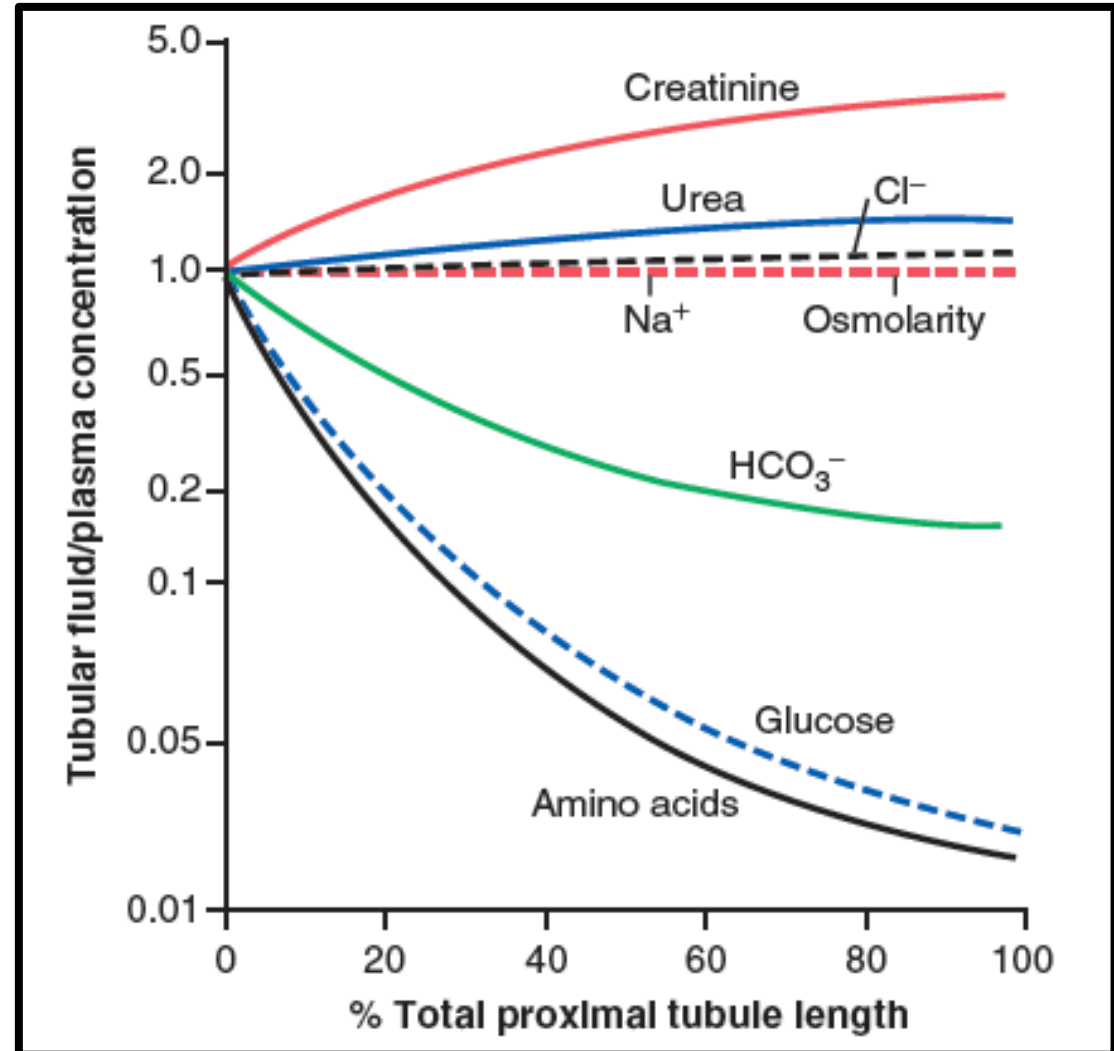
- Note that the [TF/P] ratio of Chloride (Cl^-) towards the end of PCT is slightly above 1 while that of sodium (Na^+) remains at 1.0.
- Reason: In the first half of the PCT, sodium is reabsorbed with other solutes like amino acids, glucose, bicarbonate, phosphate etc., and NOT with chloride.
- However, in the late PCT, chloride reabsorption occurs with sodium as shown in the bottom figure, but its concentration in tubular fluid is already increased compared to its plasma concentration.

[TF/P] Curves of Urea and Creatinine in PCT

[TF/P] of **Urea** goes above 1.0 because it is reabsorbed from the tubule but to a **much lesser** extent than water in PCT. Its concentration in PCT increases.

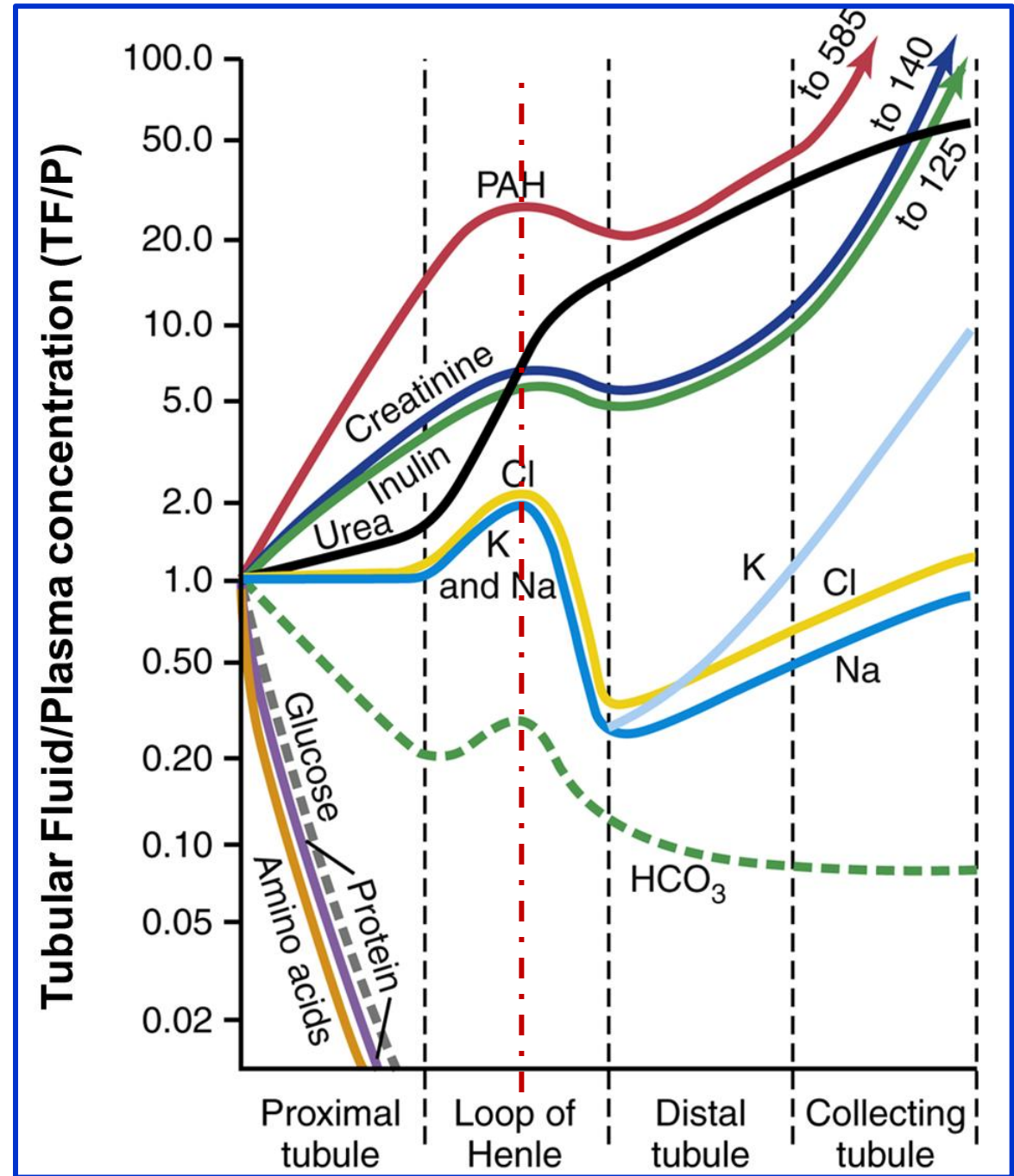
[TF/P] of **Creatinine** is much greater than 1.0 because it is **NOT reabsorbed** and is also **secreted** in the PCT. Hence its concentration in tubular fluid rapidly rises.

Tubular Fluid remains **Isosmotic** in the PCT because the amount of osmotically active particles in tubular fluid is same as that in plasma.



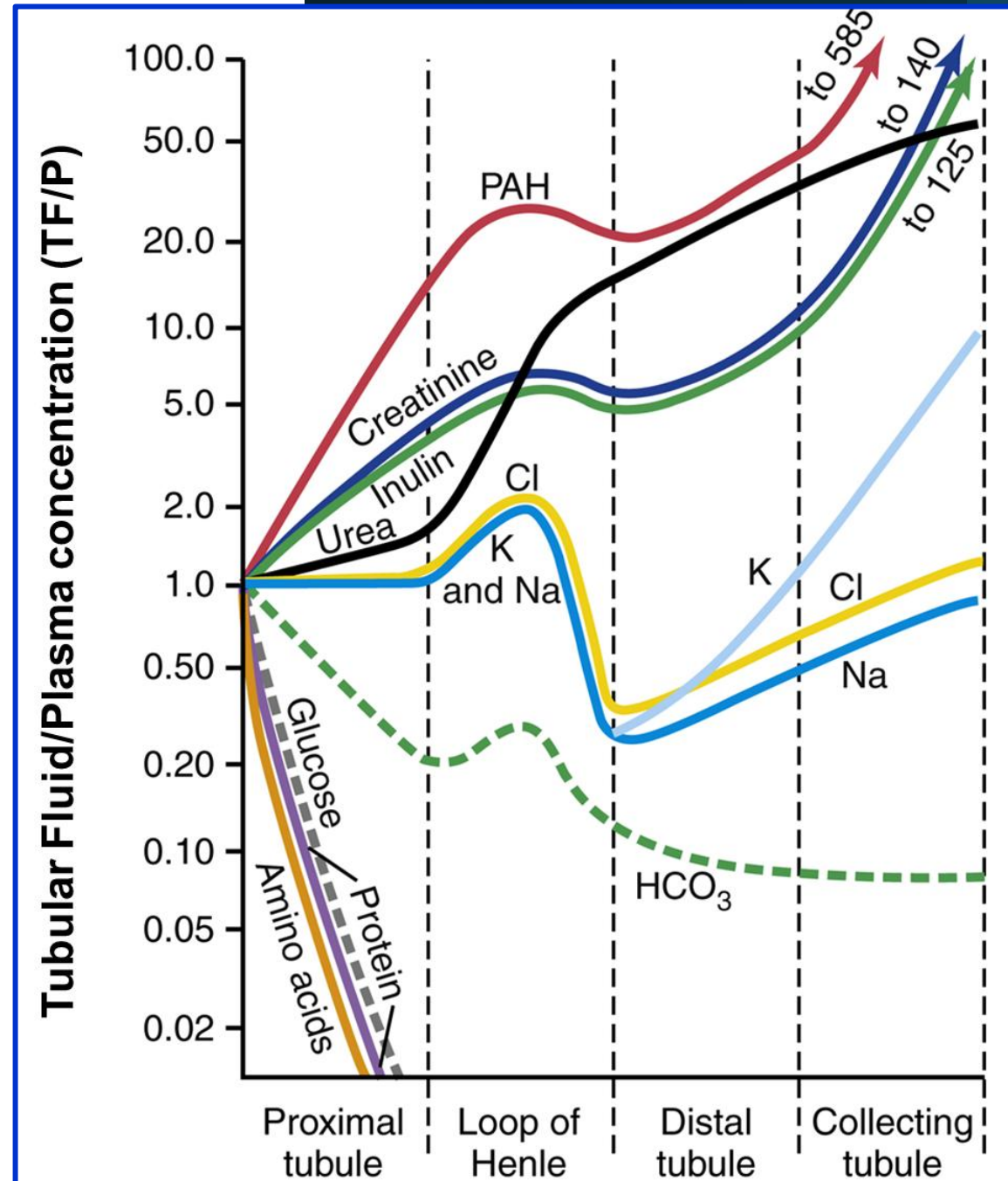
TF/P profile in Loop of Henle

- The red discontinuous line in the middle of the loop of Henle divides it into the descending limb (left half) and the ascending limb (right half).
- In descending limb, all solute concentrations **increase**. Reason: This limb is highly permeable to water (facilitates water reabsorption) but impermeable to all solutes.
- In ascending limb, all solute concentrations **decrease** because this limb is impermeable to water (no water reabsorption) but permeable to certain solutes. Since sodium, chloride and potassium are reabsorbed actively in this part of the nephron, their concentration reduces to a greater extent.
- Tubular fluid leaving the loop of Henle is **hypotonic**.

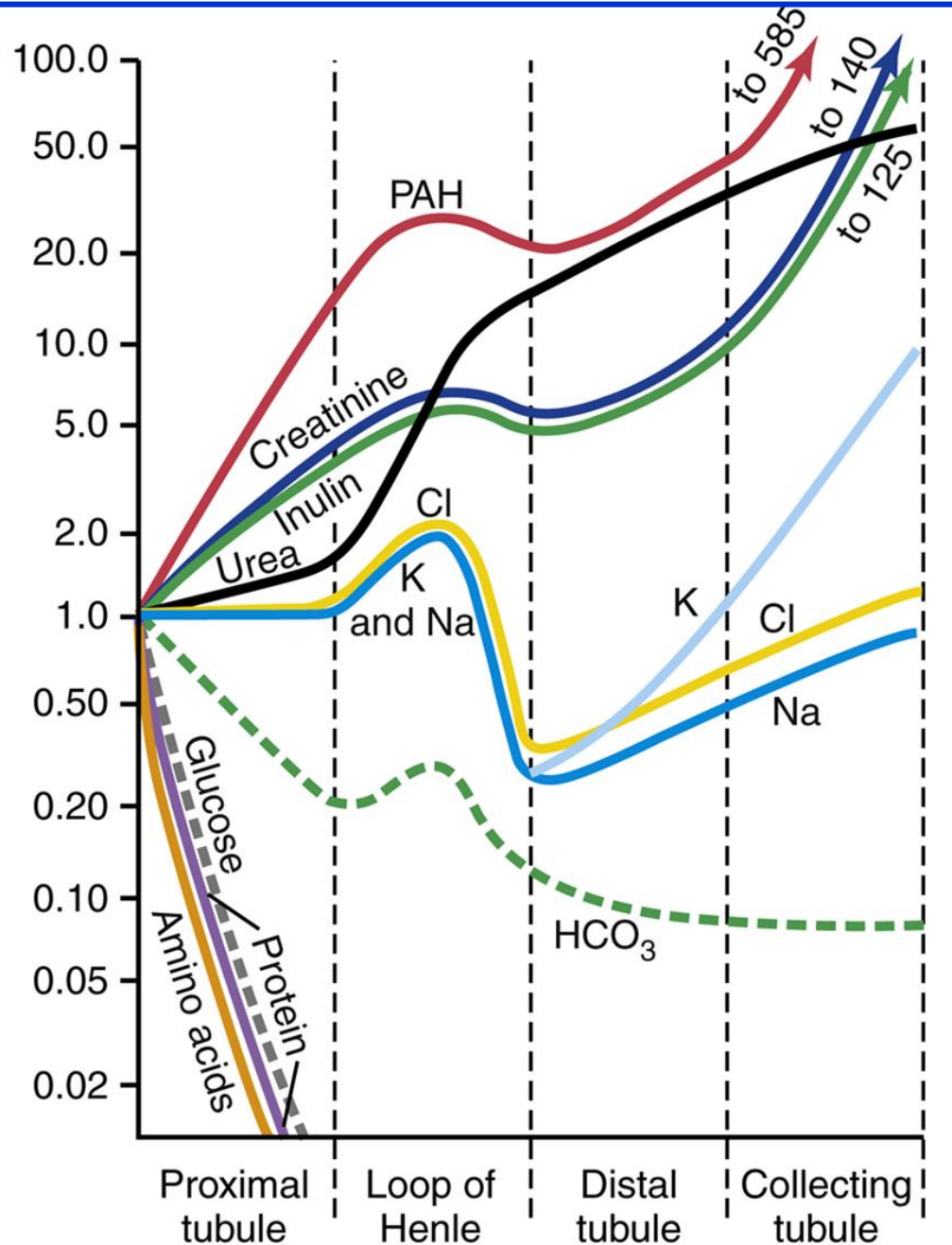


TF/P profile in DCT

- Distal tubule reabsorbs sodium, chloride, water, but secretes potassium. As water reabsorption occurs to a greater extent than Na and Cl⁻, their [TF/P] ratio is going up.
- TF/P curve of potassium is steeper than Na and Cl⁻ because it is secreted.
- Intercalated cells in DCT secrete hydrogen ions into the tubular lumen. This process makes bicarbonate reabsorption.
- Since water reabsorption continues until the osmolality of tubular fluid becomes equal to that of plasma towards the end of DCT.



Tubular Fluid/Plasma concentration (TF/P)

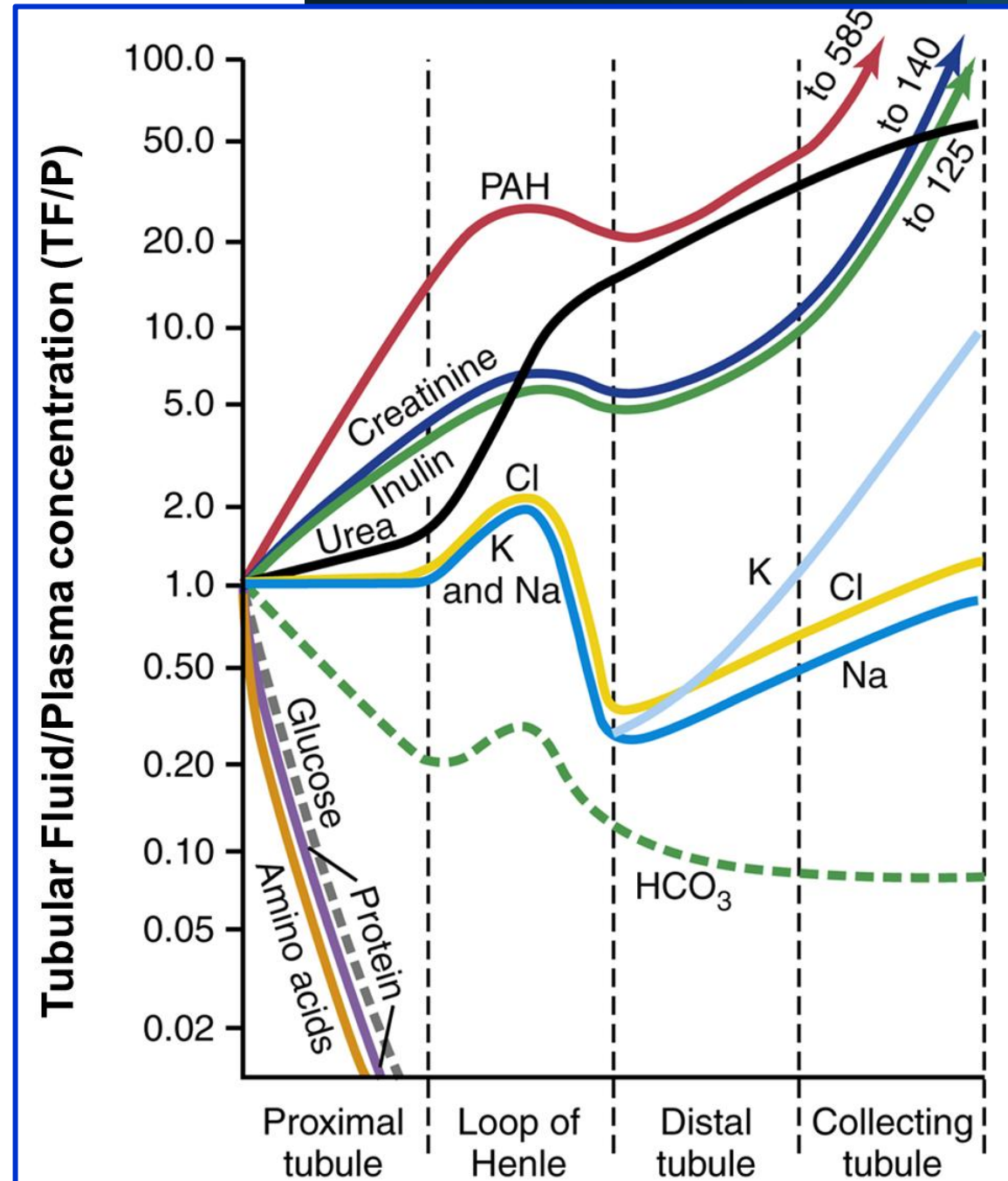


TF/P profile in Collecting tubule

- The collecting tubule (CT) is mainly concerned with **water reabsorption**; therefore, all solute concentrations and TF/P ratios of solutes increase. Here, water is being reabsorbed to a greater extent than all solutes.
- This water reabsorption in the collecting tubule is dependent on the presence of Anti diuretic hormone (ADH).
- Collecting tubules reabsorb urea (specifically in the medullary CT) under the influence of ADH. Hence, the [TF/P] curve of urea does not become steep in the inner CT.
- The tubular fluid in the CT becomes hypertonic (concentrated because of unabsorbed solutes) in the presence of ADH. It gets diluted in the Absence of ADH.

TF/P profile of Inulin, Creatinine, and PAH

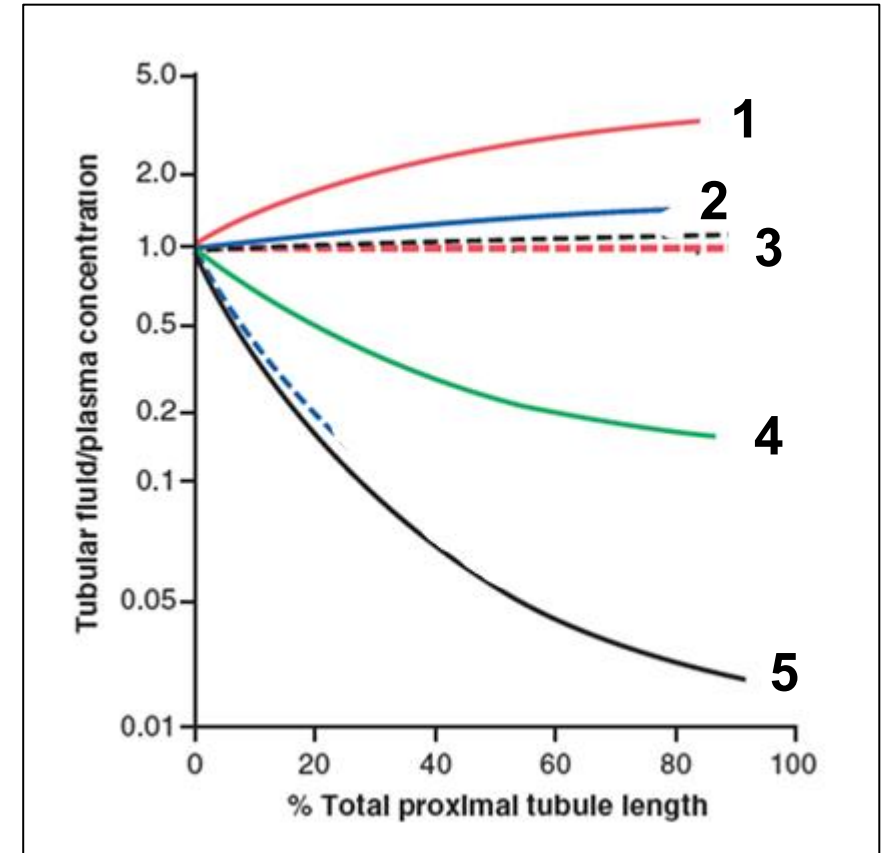
- Inulin is freely filtered, is neither secreted nor reabsorbed in the renal tubules. Therefore, its [TF/P] ratio increases as the tubular fluid moves along the different nephron segments.
- Creatinine is freely filtered, not reabsorbed but a small amount is secreted. Therefore, its [TF/P] ratio is very similar to that of Inulin but slightly above (due to secretion).
- [TF/P] curve of PAH is maximally steep because it is filtered and highly secreted in the renal tubules.



Practice Question 1

A 72-year-old man comes to the physician for a follow-up examination. He has a 15-year history of type 2 diabetes mellitus. Vital signs are within normal limits. Laboratory studies show a glucose concentration of 84 mg/dL (Normal: 80-110). A micropipette study showing [TF/P] ratio of solutes is recorded and shown in the figure. Which of the following labels best represents [TF/P] of glucose in this patient?

- A. Curve 1
- B. Curve 2
- C. Curve 3
- D. Curve 4
- E. Curve 5



Correct Answer: E. Curve 5.

Since his plasma glucose is within normal limits and below renal threshold, all glucose is reabsorbed in the PCT. Therefore, TF/P ratio is approaching zero.

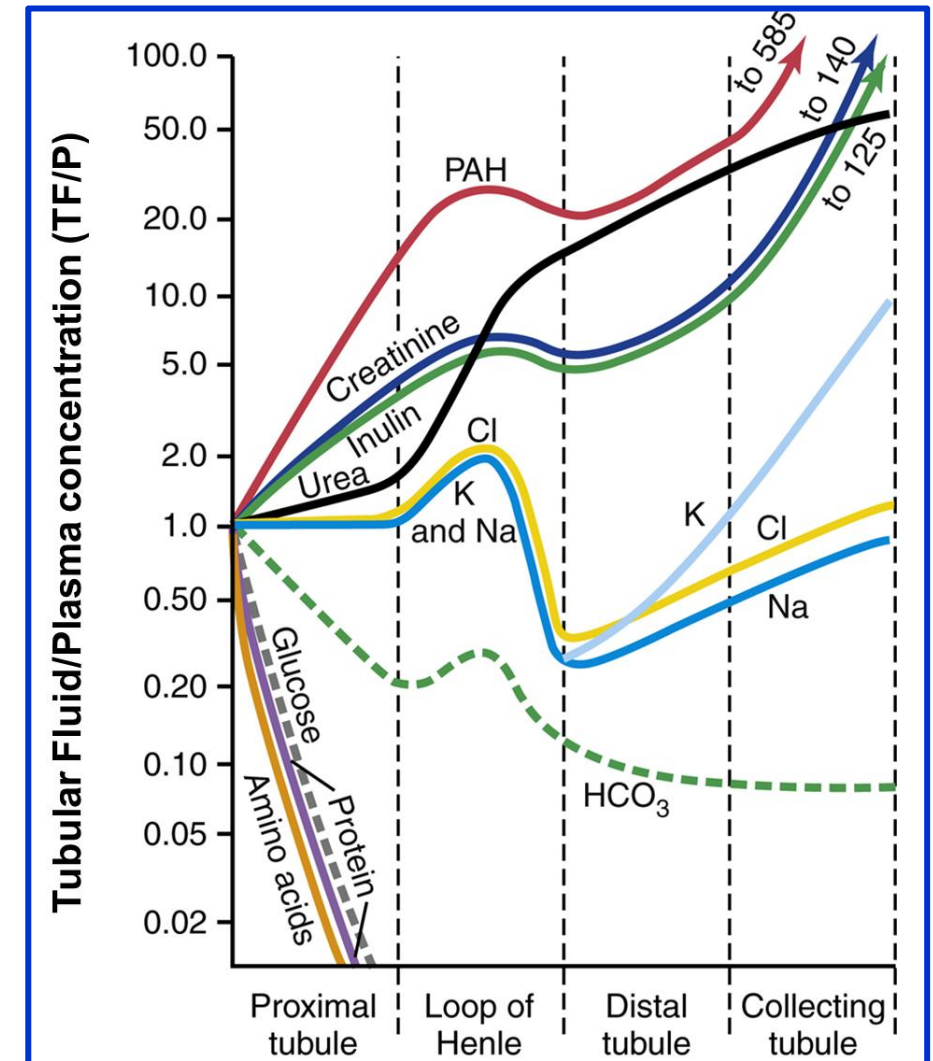
Practice Question 2

A 26-year-old man participates in a renal study. A micropipette study showing TF/P ratio of solutes in various parts of a nephron is recorded and shown in the figure. Which of the following best explains a constant rising [TF/P] ratio of creatinine in this volunteer?

- A. Creatinine is completely reabsorbed in renal tubules
- B. Creatinine is not filtered
- C. Creatinine is secreted and reabsorbed
- D. Creatinine is not reabsorbed but secreted in renal tubules
- E. Creatinine is not secreted but reabsorbed in renal tubules

Correct Answer: D.

Creatinine is freely filtered, not reabsorbed but secreted in the renal tubules.



For question and clarifications please email:
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OR

Use Office hours

References

- Medical Physiology, Principles for Clinical medicine. Rodney A Rhoades & David A. Bell 6th edition.
- Medical Physiology, Guyton & Hall 13th Edition.